

Stability

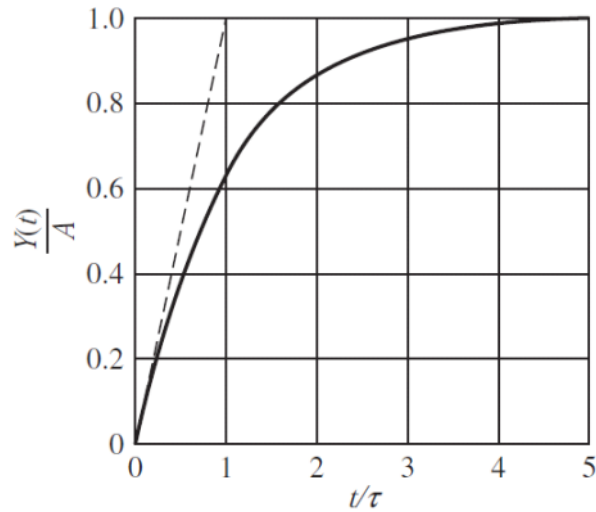


FIGURE 4-7
Response of a first-order system to a step input.

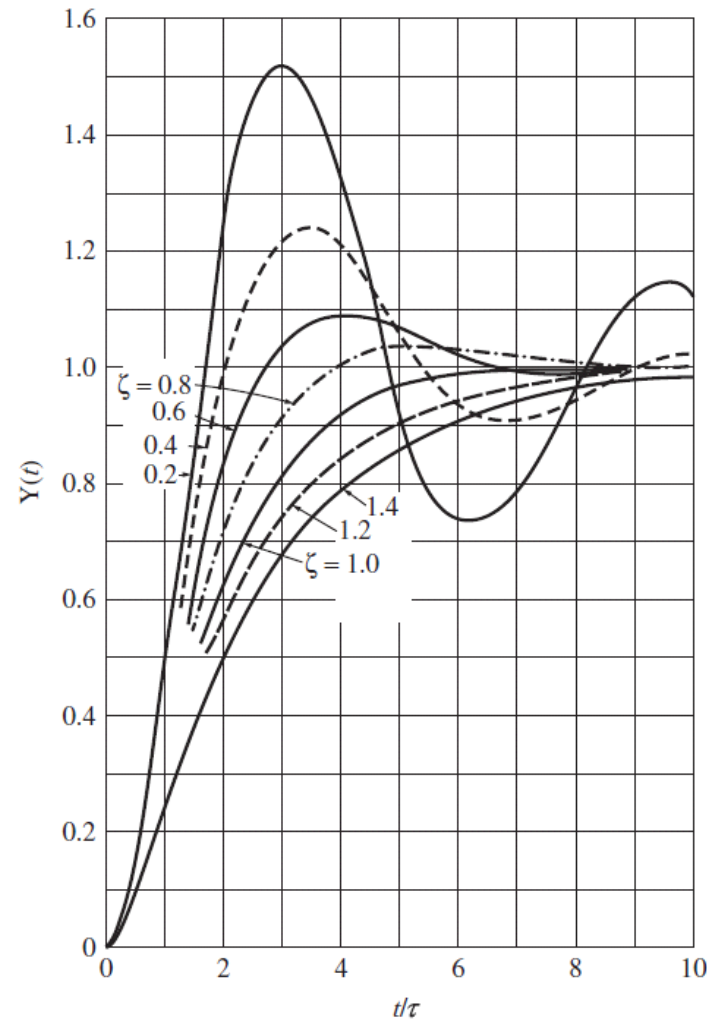


FIGURE 7-3
Response of a second-order system to a unit-step forcing function.

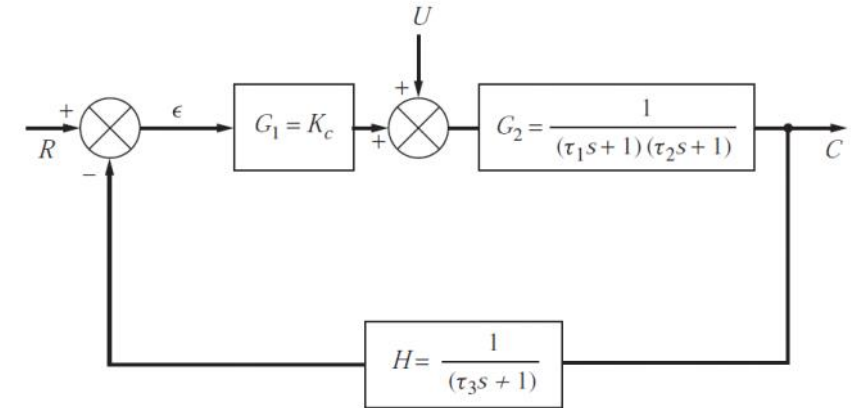
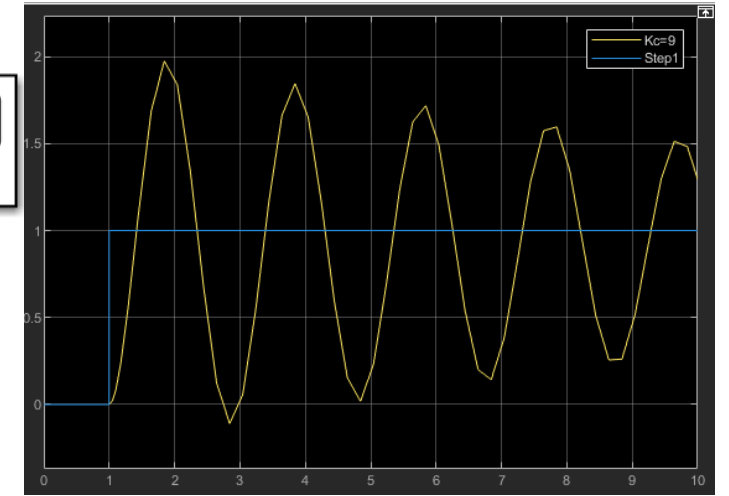
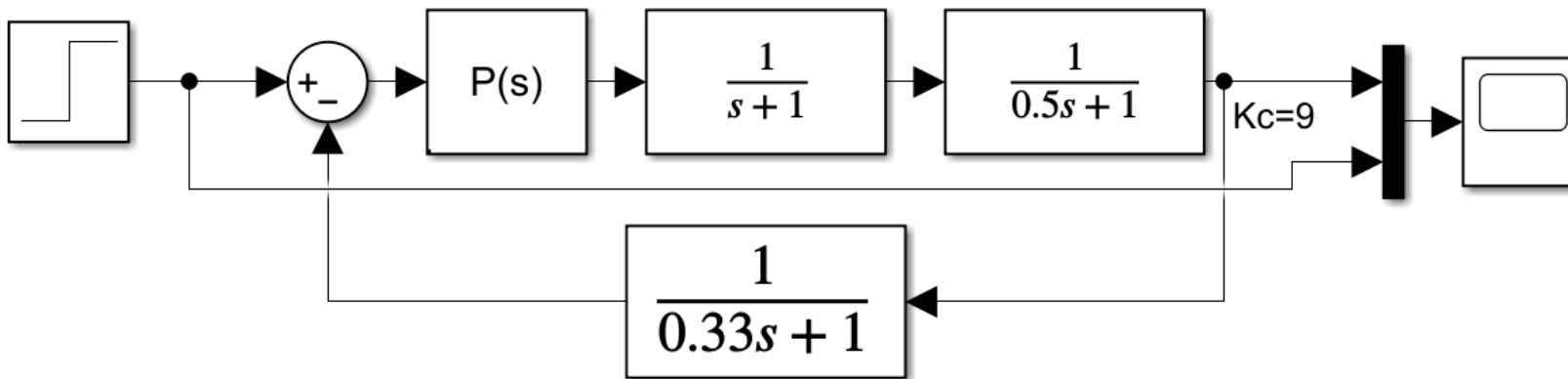


FIGURE 13-1
Third-order control system.

Overall transfer function

$$\frac{C}{R} = \frac{G_1 G_2}{1 + G_1 G_2 H}$$

$\Rightarrow$ third order



$$\frac{C}{R} = \frac{K_c / (\tau_1 s + 1)(\tau_2 s + 1)}{1 + \frac{K_c}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1)}}$$

$$= \frac{K_c}{\cancel{(\tau_1 s + 1)(\tau_2 s + 1)} + \frac{K_c}{\tau_3 s + 1}}$$

$$\frac{C}{R} = \frac{K_c (\tau_3 s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1) + K_c}$$

$\Rightarrow$
for step
change
 $R = \frac{1}{s}$

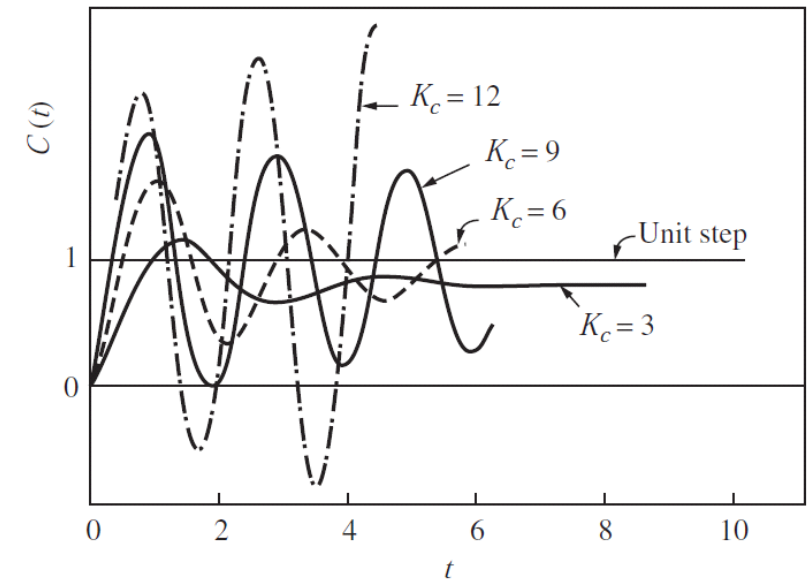
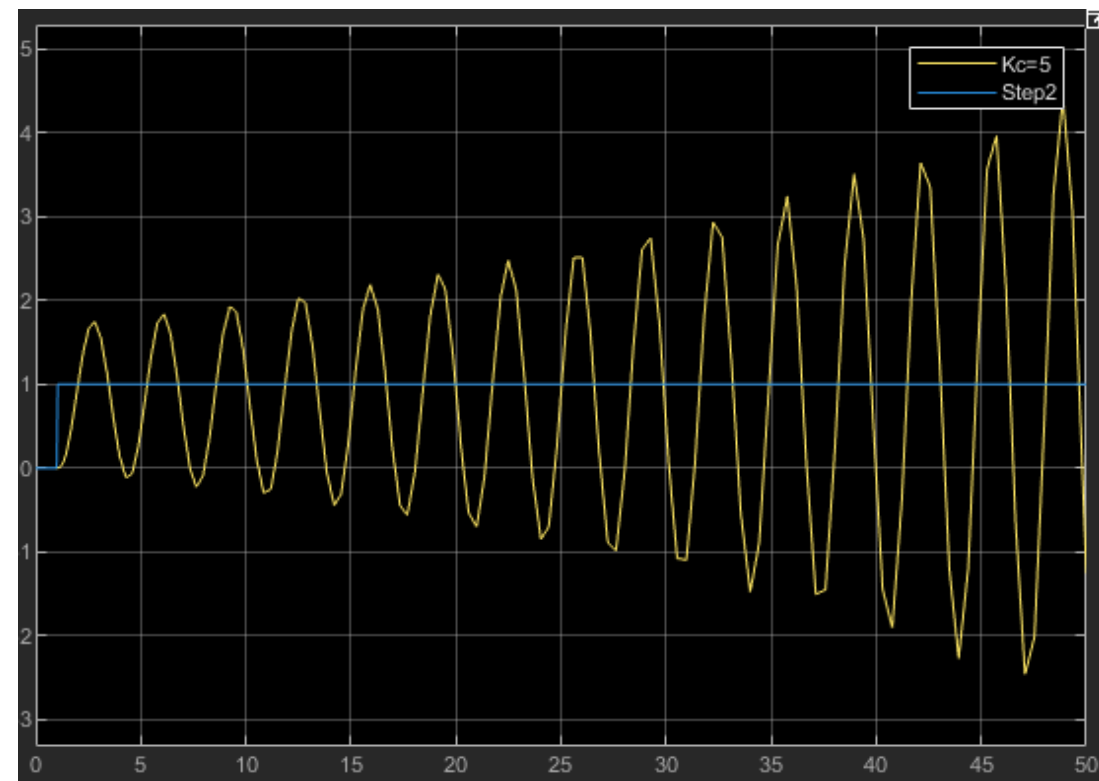
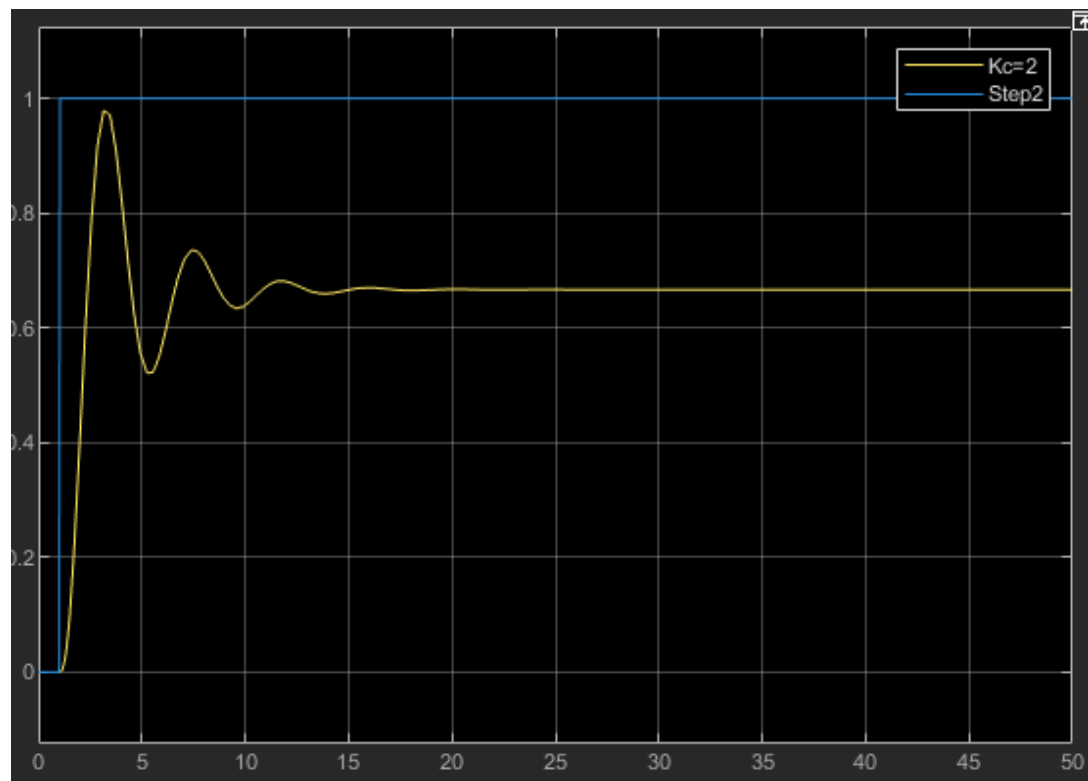
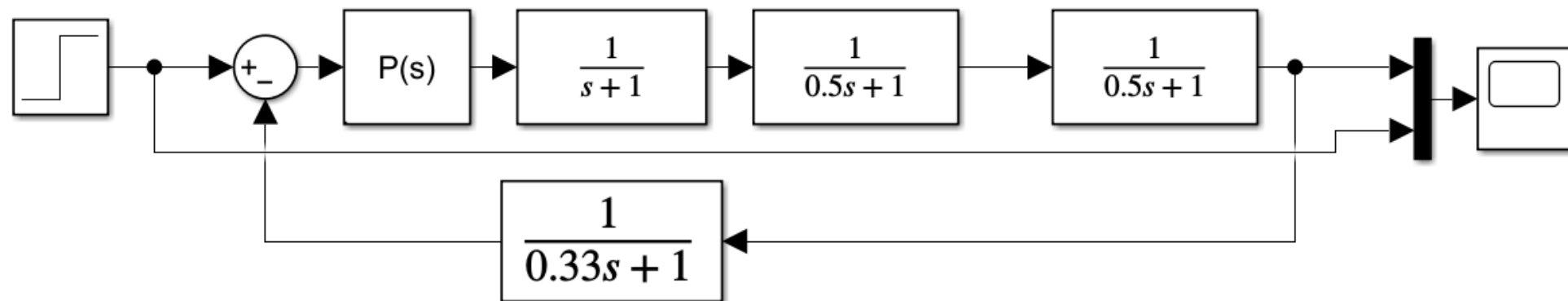


FIGURE 13-2
Response of control system of Fig. 13-1 for a unit-step change in set point.



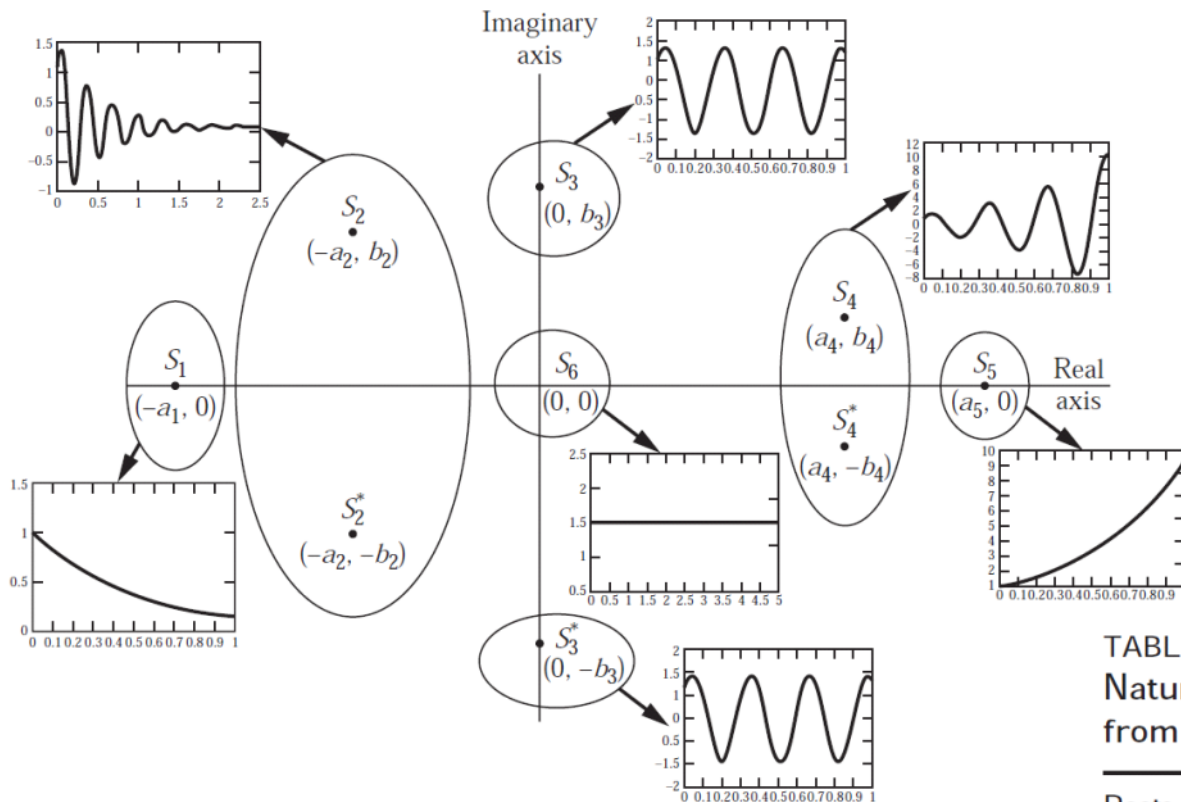


FIGURE 3-1
Location of typical roots of characteristic equation.

TABLE 3.1
Nature of terms in the solution $x(t)$ based on roots in the denominator of $X(s)$ from Fig. 3-1

Roots in denominator of $X(s)$	Terms in $x(t)$ for $t > 0$
s_1	$C_1 e^{-a_1 t}$
s_2, s_2^*	$e^{-a_2 t} (C_1 \cos b_2 t + C_2 \sin b_2 t)$
s_3, s_3^*	$C_1 \cos b_3 t + C_2 \sin b_3 t$
s_4, s_4^*	$e^{a_4 t} (C_1 \cos b_4 t + C_2 \sin b_4 t)$
s_5	$C_1 e^{a_5 t}$
s_6	C_1

Stability - Defined

↳ For Linear Systems

Bounded Inputs

→ Step

→ Sinusoidal

Unbounded Input

→ $f(t) = t$

" A system exhibiting unbounded response to a bounded input is unstable. "

" Stable system has bounded input with bounded output". (BIBO)

Stability Criterion

→ Considering a control system of following form

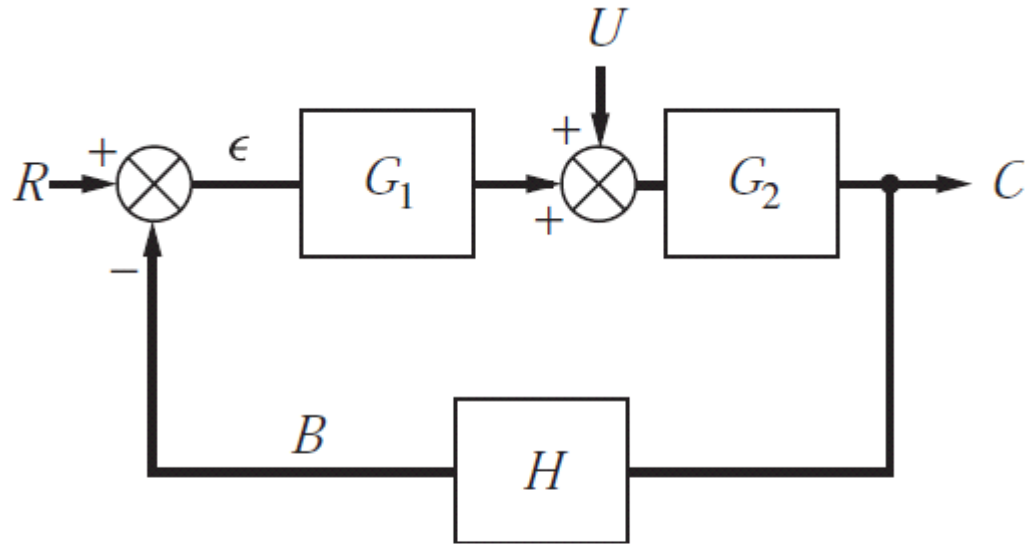


FIGURE 13-3

Basic single-loop control system.

Characteristic Equation

$$C = \frac{G_1 G_2}{1 + G_1 G_2 H} R + \frac{G_2}{1 + G_1 G_2 H} U$$

$G = \text{Open loop transfer func.}$

$$C = \frac{G_1 G_2}{1 + G} R + \frac{G_2}{1 + G} U$$

Testing the response with a bounded input

→ Unit step change in set point

$$\Rightarrow C(s) = \frac{G_1 G_2}{1 + G} \frac{1}{s}$$

$$C(s) = \frac{G_1 G_2}{1+G} \frac{1}{s}$$

$\Rightarrow$ Roots of Characteristic Eqn

$$1+G=0$$

$\hookrightarrow$ Roots $\Rightarrow \gamma_1, \gamma_2, \gamma_3$

$$1+G = (s-\gamma_1)(s-\gamma_2)(s-\gamma_3)$$

$$C(s) = \frac{G_1 G_2}{1+G} \frac{1}{s} = \frac{G_1 G_2 F(s)}{s(s-\gamma_1)(s-\gamma_2)(s-\gamma_3)}$$

$F(s) \rightarrow$ arises due to rearrangement of terms

Continuing $\Rightarrow \frac{C}{R} = \frac{K_c(\tau_3 s + 1)}{(\tau_1 s + 1)(\tau_2 s + 1)(\tau_3 s + 1) + K_c}$
with $R = \frac{1}{s}$

$$C = \frac{K_c(\tau_3 s + 1)}{s \left[\tau_1 \tau_2 \tau_3 s^3 + (\tau_1 \tau_2 + \tau_1 \tau_3 + \tau_2 \tau_3) s^2 + (\tau_1 + \tau_2 + \tau_3) s + (1 + K_c) \right]}$$

Will have 3 roots
 $\gamma_1, \gamma_2, \gamma_3$

with general form

$$\tau_1 \tau_2 \tau_3 \left[s^3 + \frac{(\tau_1 \tau_2 + \tau_1 \tau_3 + \tau_2 \tau_3)}{\tau_1 \tau_2 \tau_3} s^2 + \frac{(\tau_1 + \tau_2 + \tau_3)}{\tau_1 \tau_2 \tau_3} s + \frac{(1 + K_c)}{\tau_1 \tau_2 \tau_3} \right]$$

$$C = \frac{K_c(\tau_3 s + 1)}{\tau_1 \tau_2 \tau_3 (s(s-\gamma_1)(s-\gamma_2)(s-\gamma_3))}$$

$$\Rightarrow F(s) = \frac{K_c(\tau_3 s + 1)}{\tau_1 \tau_2 \tau_3}$$

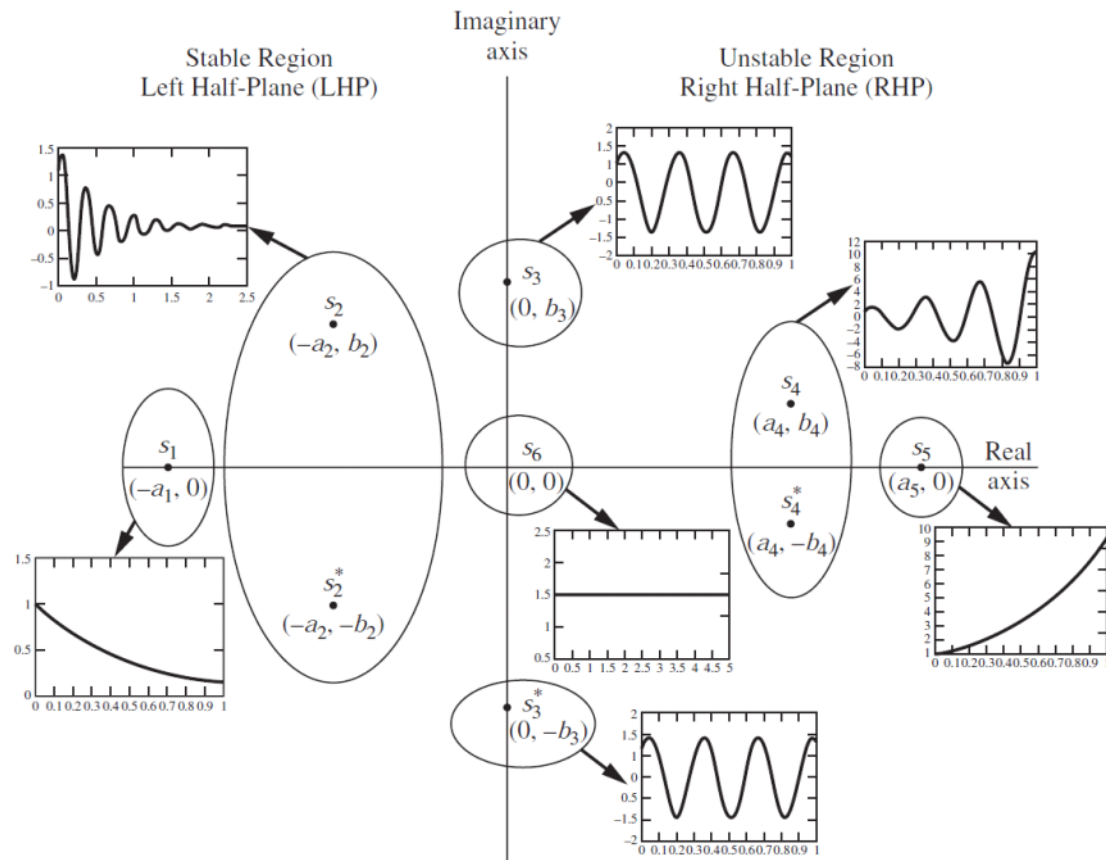


FIGURE 13-4
Stability of typical roots of the characteristic equation.

Therefore, the definition of *stability for linear systems* may be translated to the following criterion: A linear control system is unstable if any roots of its characteristic equation are on, or to the right of, the imaginary axis. Otherwise the system is stable.

TABLE 3.1
Nature of terms in the solution $x(t)$ based on roots in the denominator of $X(s)$ from Fig. 3-1

Roots in denominator of $X(s)$	Terms in $x(t)$ for $t > 0$
s_1	$C_1 e^{-a_1 t}$
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s_3, s_3^*	$C_1 \cos b_3 t + C_2 \sin b_3 t$
s_4, s_4^*	$e^{a_4 t} (C_1 \cos b_4 t + C_2 \sin b_4 t)$
s_5	$C_1 e^{a_5 t}$
s_6	C_1

Determination of stability is based on roots of characteristic eqn which is based on the open loop transfer function.

Example 13.1

For a control system with transfer functions

$$G_1 = 10 \frac{0.5s+1}{s} \quad \text{Controller}$$

$$G_2 = \frac{1}{2s+1} \quad \text{Tank/Process}$$

$$H = 1 \quad \text{Measuring Element}$$

$$G = G_1 G_2 H = \frac{10(0.5s+1)}{s(2s+1)}$$

Finding Roots of Characteristic eqn

$$1+G=0$$

$$1 + \frac{10(0.5s+1)}{s(2s+1)} = 0$$

$$\frac{2s^2 + s + 5s + 10}{2s^2 + s} = 0$$

$$2s^2 + 6s + 10 = 0$$

$$s^2 + 3s + 5 = 0$$

$$s = -\frac{3}{2} \pm \frac{\sqrt{9-20}}{2}$$

$$s_1 = -\frac{3}{2} + i \frac{\sqrt{11}}{2} \quad ; \quad s_2 = -\frac{3}{2} - i \frac{\sqrt{11}}{2}$$

First term $-\frac{3}{2}$ is -ive $\Rightarrow$ Stable system

Routh test for Stability

→ Used to determine how many roots of the characteristic eqn have positive real parts

↳ If no positive real parts $\Rightarrow$ stable system

↳ Used only for polynomial eqns

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

Coefficients need to be positive for roots to lie in left plane

↳ system may still be stable or unstable

Procedure for determining stability

→ Routh array

Row			
1	a_0	a_2	a_4
2	a_1	a_3	a_5
3	b_1	b_2	
4	c_1	c_2	
5	d_1		

⇓
All positive $\Rightarrow$ stable system

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

$$c_1 = \frac{b_1 a_3 - b_2 a_1}{b_1}$$

⋮

Example 13-2

For characteristic Eqn

$$s^4 + 3s^3 + 5s^2 + 4s + 2 = 0$$

Determine stability by Routh test

Row			
1	1	5	2
2	3	4	
3	$\frac{3 \times 5 - 1 \times 4}{3} = \frac{11}{3}$	$\frac{3 \times 2 - 1 \times 0}{3} = \frac{6}{3}$	
4	$\frac{26}{11}$	0	
5	2		

In first column
 $\Rightarrow$ No sign changes $\rightarrow$ All +ive
 $\Rightarrow$ Stable system

Example 13.3

for system in 13-1

Using $\tau_1 = 1$, $\tau_2 = \frac{1}{2}$, $\tau_3 = \frac{1}{3}$

Determine values of K_c for which system will be stable

⇒ Characteristic Eqn

$$1 + G = 0$$

$$1 + \frac{K_c}{(s+1)(s/2+1)(s/3+1)} = 0$$

$$(s+1) \left(\frac{s}{2} + 1 \right) \left(\frac{s}{3} + 1 \right) + K_c = 0$$

$$\left(\frac{s^2}{2} + s + \frac{s}{2} + 1 \right) \left(\frac{s}{3} + 1 \right) + K_c = 0$$

$$\frac{s^3}{6} + \frac{s^2}{3} + \frac{s^2}{6} + \frac{s}{3} + \frac{s^2}{2} + s + \frac{s}{2} + 1 + K_c = 0$$

$$\frac{s^3}{6} + s^2 \left(\frac{1}{3} + \frac{1}{6} + \frac{1}{2} \right) + s \left(\frac{1}{3} + 1 + \frac{1}{2} \right) + (1 + K_c) = 0$$

$$s^3 + s^2 6 \left(\frac{2+1+3}{6} \right) + s 6 \left(\frac{2+6+3}{6} \right) + 6(1+K_c) = 0$$

$$s^3 + 6s^2 + 11s + 6(1+K_c) = 0$$

Routh array

a) Value of K_c for stability

Row		
1	1	11
2	6	$6(1+K_c)$
3	$\frac{6 \times 11 - 6(1+K_c)}{6}$ $= 10 + K_c$	
4	$\frac{(10+K_c)(6(1+K_c)) - 0}{10+K_c}$ $= 6(1+K_c)$	

For stability $\Downarrow$ All positive

i.e. $10 - K_c > 0$

$\Rightarrow K_c < 10 \Rightarrow$ Unstable over 10

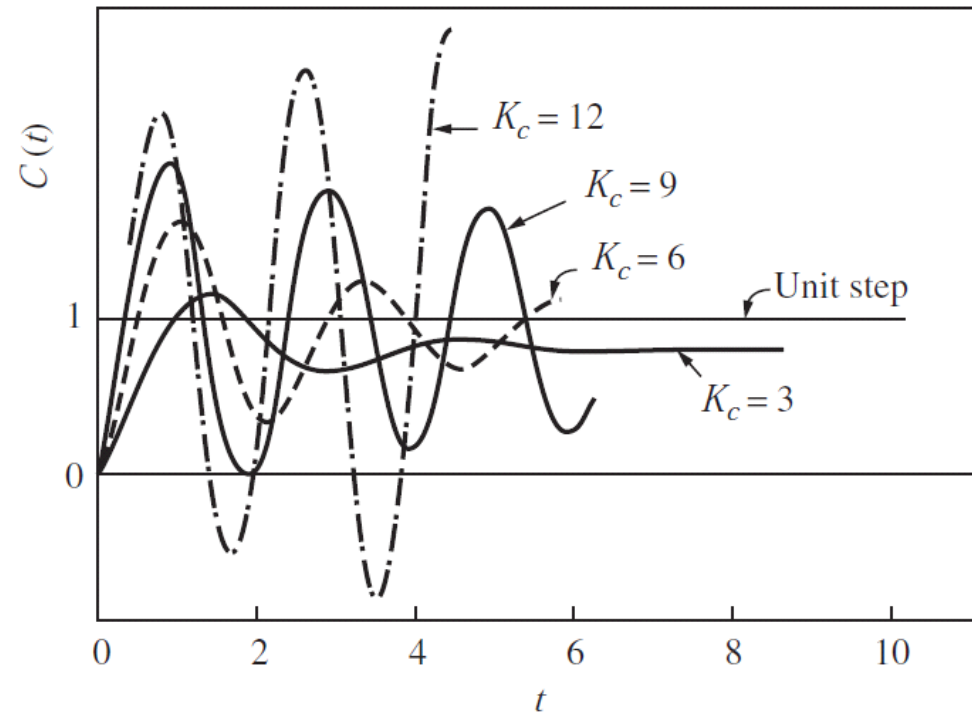


FIGURE 13-2

Response of control system of Fig. 13-1 for a unit-step change in set point.

b) For value of K_c where system is on threshold of instability
Determine roots of characteristic eqn using Theorem 13.3

Imaginary roots solved using

$$Cs^2 + D = 0$$

C and $D \Rightarrow$ elements of $n-1$ row

Threshold of instability @ Row 3 $\Rightarrow 10 - K_c$

$(n-1)$ row $\Rightarrow$ values 6 and $6(1 + K_c)$

@ $K_c = 10$

$$C = 6, D = 6(1 + 10) = 66$$

$$\Rightarrow Cs^2 + D = 0$$

$$6s^2 + 66 = 0$$

$$6s^2 = -66$$

$$s^2 = -11$$

$$s = \sqrt{-11} = \pm i\sqrt{11}$$

Imaginary roots located

@ $-\sqrt{11}$ and $\sqrt{11}$

$\Rightarrow$ Two roots found

Third root found using factoring

$$(s-s_1)(s-s_2)(s-s_3)=0$$

$$\downarrow$$
$$i\sqrt{11}$$

$$\downarrow$$
$$-i\sqrt{11}$$

$$(s-i\sqrt{11})(s+i\sqrt{11})(s-s_3)=0$$

$$(s^2 + \cancel{s i\sqrt{11}} - \cancel{s i\sqrt{11}} - \underbrace{i^2}_{-1} \underbrace{\sqrt{11}^2}_{11})(s-s_3)=0$$

$$(s^2 + 11)(s-s_3)=0$$

$$s^3 - s^2 s_3 + 11s - 11s_3 = 0$$

Compare with

$$s^3 + 6s^2 + 11s + 6(1+K_c)=0$$

Shows

$$s_3 = -6$$

All roots

$$s_1 = i\sqrt{11}$$

$$s_2 = -i\sqrt{11}$$

$$s_3 = -6$$

THEOREMS OF THE ROUTH TEST

Theorem 13.1. The necessary and sufficient condition for all the roots of the characteristic equation [Eq. (13.9)] to have negative real parts (stable system) is that all elements of the first column of the Routh array (a_0, a_1, b_1, c_1 , etc.) be positive and nonzero.

Theorem 13.2. If some of the elements in the first column are negative, the number of roots with a positive real part (in the right half-plane) is equal to the number of sign changes in the first column.

Theorem 13.3. If *one* pair of roots is on the imaginary axis, equidistant from the origin, and all other roots are in the left half-plane, then all the elements of the n th row will vanish and none of the elements of the preceding row will vanish. The location of the pair of imaginary roots can be found by solving the equation

$$Cs^2 + D = 0 \quad (13.10)$$

where the coefficients C and D are the elements of the array in the $(n - 1)$ st row as read from left to right, respectively.

Example 13.4

PI controller used

$$\tau_1 = 1, \tau_2 = \frac{1}{2}, \tau_3 = \frac{1}{3}$$

$$K_c = 5, \tau_I = 0.25$$

Determine stability

Characteristic eqn for PI control

$$1 + G = 0$$

$$1 + \frac{\left(\frac{K_c}{\tau_1 \tau_2 \tau_3}\right) (\tau_I s + 1)}{\tau_I s \left(s + \frac{1}{\tau_1}\right) \left(s + \frac{1}{\tau_2}\right) \left(s + \frac{1}{\tau_3}\right)} = 0$$

$$1 + \frac{\left(\frac{5}{1 \cdot \frac{1}{2} \cdot \frac{1}{3}}\right) \left(\frac{s}{4} + 1\right)}{\frac{s}{4} (s+1) (s+2) (s+3)} = 0$$

$$\Rightarrow 1 + \frac{\frac{30}{4} (s+4)}{\frac{s}{4} (s+1) (s+2) (s+3)} = 0$$

$$s (s+1) (s+2) (s+3) + 30 (s+4) = 0$$

$$s (s^2 + 2s + s + 2) (s+3) + 30s + 120 = 0$$

$$s \left(\begin{matrix} s^3 + 2s^2 + s^2 + 2s \\ + 3s^2 + 6s + 3s + 6 \end{matrix} \right) + 30s + 120 = 0$$

$$s^4 + 2s^3 + s^3 + 2s^2 + 30s + 120 = 0$$

$$+ 3s^3 + 6s^2 + 3s^2 + 6s$$

$$s^4 + s^3 (2+1+3) + s^2 (2+6+3) + s (6+30) + 120 = 0$$

$$s^4 + 6s^3 + 11s^2 + 36s + 120 = 0$$

Row 1	1	11	120
2	6	36	
3	$\frac{6 \times 11 - 36}{6}$ = 5	$\frac{6 \times 120 - 0}{6}$ = 120	
4	$\frac{5 \times 36 - 6 \times 120}{5}$ = -108		
5	$\frac{-108 \times 120 - 0}{-108}$ = 120		

There are two sign changes in first column $\Rightarrow$ 2 roots have +ive real parts
 $\Rightarrow$ System unstable for $K_c = 5$